TEXTBOOK EXERCISES

❖ I. Choose the correct answer.

1. The separation of denser particles from lighter particles done by rotation at high speed is called _____
   a) Filtration b) sedimentation
   c) decantation d) centrifugation
2. Among the following _____ is a mixture
   a) Common Salt b) Juice
   c) Carbon dioxide d) Pure Silver
3. When we mix a drop of ink in water we get a _____
   a) Heterogeneous Mixture b) Compound
   c) Homogeneous Mixture d) Suspension
4. _____ is essential to perform separation by solvent extraction method.
   a) Separating funnel b) filter paper
   b) centrifuge machine d) sieve
5. _____ has the same properties throughout the sample
   a) Pure substance b) Mixture
   c) Colloid d) Suspension

II. State whether true or false. If false, correct the statement.

1. Oil and water are immiscible with each other.
2. A compound cannot be broken into simpler substances chemically.
3. Liquid – liquid colloids are called gel.
4. Buttermilk is an example of heterogeneous mixture.
5. Aspirin is composed of 60% Carbon, 4.5% Hydrogen and 35.5% Oxygen by mass. Aspirin is a mixture.

III. Match the following.

Element	Settles down on standing
Compound	Impure substance
Colloid	Made up of molecules
Suspension	Pure substance
Mixture	Made up of atoms

IV. Fill in the blanks.

1. A _____ mixture has no distinguishable boundary between its components.
2. An example of a substance that sublimes is _____
3. Alcohol can be separated from water by _____
4. In petroleum refining, the method of separation used is _____
5. Chromatography is based on the principle of _____

V. Answer very briefly.

1. Diffentiate between absorption and adsorption.
2. Define Sublimation.
3. A few drops of 'Dettol' when added to water the mixture turns turbid. Why?
4. Name the apparatus that you will use to separate the components of mixtures containing two, i. miscible liquids, ii. immiscible liquids.
5 Name the components in each of the following mixtures.
   i. Ice cream ii. Lemonade
   iii. Air iv. Soil

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